

Yes. If the objective is “do not miss any serious model that could influence SatQuery’s architecture”, we should not make a tiny Top-10 list. The landscape spans **RS-VLMs, EO foundation models, grounding, segmentation, temporal models, multimodal sensor models, geospatial foundation models, and reasoning/autentic systems**.

I cross-checked current 2026 landscape sources and benchmarks. VRSBench alone compares several generations of RS-VLMs, while newer 2026 evaluations now include models such as Earth-OneVision, RSThinker, EarthDial and SkySenseGPT.

Complete SatQuery Model Universe

I would organize the models into **10 capability families** rather than pretending that one leaderboard can tell us which model is "best."

Tier 1 — Models we MUST deeply study

These are the ones most likely to influence the eventual SatQuery core.

Model	Main superpower	SatQuery importance
Earth-OneVision	Broad multimodal RS-MLLM + many sensors/tasks	★★★★★
RemoteReasoner	Unified geospatial reasoning/workflow	★★★★★
TerraMind	Any-to-any multimodal Earth reasoning	★★★★★
SkySense V2	Unified multimodal EO foundation model	★★★★★
SkySense++	Large-scale multimodal/few-shot EO representation	★★★★★
MIGRANT	Multi-image grounding/reasoning	★★★★★
GeoGround	Grounding + HBB/OBB/masks	★★★★★
Think2Seg-RS	VLM reasoning → segmentation	★★★★★
VHM	Honest/versatile RS-VLM	★★★★★

EarthGPT / EarthGPT-X	Optical + SAR + IR multimodal understanding	★★★★★
SkyEyeGPT	Broad RS instruction-following	★★★★
GeoChat	Grounded RS conversation	★★★★
LHRS-Bot	RS + geographic/VGI knowledge	★★★★
SATGround	Advanced spatial grounding	★★★★
Prithvi-EO-2.0	Temporal + multispectral EO foundation model	★★★★★
AlphaEarth Foundations	Massive-scale Earth embeddings	★★★★★



1. General RS Vision-Language Models

These answer:

"Can I talk to a satellite image intelligently?"

Core models

1. **GeoChat**
2. **SkyEyeGPT**
3. **EarthGPT**
4. **EarthGPT-X**
5. **VHM**
6. **LHRS-Bot**
7. **RSGPT**
8. **SkySenseGPT**
9. **GeoLLaVA**
10. **GeoLLaVA-8K**
11. **EarthDial**
12. **GeoPix**
13. **RS-LLaVA**
14. **Remote Sensing ChatGPT**
15. **RemoteGPT**
16. **GeoChat-2 / later GeoChat variants where applicable**
17. **Earth-OneVision**

18. **RSThinker**
19. **GeoVLM-R1**
20. **RemoteReasoner**

The broader EO-VLM ecosystem includes these families and explicitly separates foundation models, captioning, retrieval, grounding and VQA systems.

What they collectively give SatQuery

- image understanding
 - VQA
 - captioning
 - object recognition
 - counting
 - spatial relations
 - conversation
 - instruction following
 - grounding
 - increasingly multimodal reasoning.
-

2. Visual Grounding / Localization Models

This family answers:

"WHERE exactly is the thing you're talking about?"

Must study

1. **GeoGround**
2. **SATGround**
3. **SkyEyeGPT**
4. **GeoChat**
5. **EarthGPT**
6. **LHRS-Bot**
7. **Earth-OneVision**
8. **RSThinker**
9. **RemoteReasoner**
10. **RSVG-specific specialist models**
11. **DIOR-RSVG models**

12. **SkySenseGPT**
13. **GeoPix**
14. **Grounding DINO adaptations for RS**
15. **SAM/SAM2-based RS grounding pipelines**

This area is changing quickly: a current RSVG leaderboard lists a specialist at **84.48 P@0.5**, while Earth-OneVision is at **65.06** and RSThinker at **64.0**, illustrating why a general-purpose VLM and a specialized grounding model should not be treated as the same thing.

SatQuery lesson

Potential outputs:

object

- bounding box
 - oriented bounding box
 - polygon
 - coordinates
 - confidence
-

3. Segmentation / Pixel-Level Intelligence

These answer:

"Which exact pixels belong to this object/region?"

Important systems

1. **Think2Seg-RS**
2. **GeoGround**
3. **GeoPix**
4. **Popeye**
5. **SAM**
6. **SAM 2**
7. **Grounded-SAM**
8. **Grounding DINO + SAM**
9. **Remote-SAM**
10. **RS-SAM adaptations**
11. **Prithvi-based segmentation systems**
12. **SatMAE-based segmentation**

13. **DOFA-based segmentation**
14. **Clay-based segmentation**
15. **RemoteReasoner**
16. **TerraMind task adapters**

The 2026 segmentation literature now broadly clusters around Transformer models, Mamba/state-space models, prompt-driven SAM-style systems, and multimodal/self-supervised foundation models.

SatQuery lesson

Don't make the language model predict masks itself.

Better:

Language reasoning



"find all damaged buildings"



Grounding



SAM / segmentation specialist



pixel masks



4. Earth Observation Foundation Models

These don't primarily "chat."

Their superpower is:

learning representations of Earth.

Critical models

1. **Prithvi-EO**
2. **Prithvi-EO-2.0**
3. **SatMAE**
4. **SatMAE++**
5. **SatlasPretrain**

6. **Scale-MAE**
7. **MAE-Geo**
8. **DOFA**
9. **Clay**
10. **Clay Foundation Model**
11. **SkySense**
12. **SkySense++**
13. **SkySense V2**
14. **TerraMind**
15. **AlphaEarth Foundations**
16. **RemoteCLIP**
17. **GeoRSCLIP**
18. **CSP / cross-sensor pretraining families**
19. **RingMo**
20. **GFM / geospatial foundation model families**
21. **Tessera**
22. **SatVision**
23. **SSL4EO-type foundation models**

A recent 2026 review of geospatial foundation models specifically discusses families such as **SatMAE**, **Scale-MAE**, **Prithvi**, **Clay**, **Tessera** and **DOFA**, so these shouldn't be ignored just because they aren't conversational models.

5. Multisensor / Multimodal Earth Models

This is probably one of the most important groups for your ultimate architecture.

Models

1. **EarthGPT**
2. **EarthGPT-X**
3. **TerraMind**
4. **SkySense**
5. **SkySense++**
6. **SkySense V2**
7. **Earth-OneVision**
8. **Prithvi-EO-2.0**
9. **DOFA**
10. **Clay**

11. **SatlasPretrain**
12. **AlphaEarth Foundations**
13. **CROMA**
14. **RemoteCLIP**
15. **GeoRSCLIP**
16. **RingMo**
17. **SSL4EO**
18. **AnySat**
19. **M3FM / multimodal geospatial foundation-model families**

Sensor combinations to investigate

RGB

+

Multispectral

+

SAR

+

Infrared

+

Temporal imagery

+

DEM

+

Geographic coordinates

+

Maps

TerraMind is particularly important because its architecture explicitly targets **multiple geospatial modalities and any-to-any generation**, rather than simply concatenating RGB and SAR features.



6. Temporal / Multi-Image Intelligence

This category is **absolutely essential** for a serious SatQuery.

The question changes from:

"What is here?"

to:

"What changed here?"

Models / systems

1. **MIGRANT**
2. **Prithvi-EO**
3. **Prithvi-EO-2.0**
4. **SkySense**
5. **SkySense++**
6. **SkySense V2**
7. **TerraMind**
8. **AlphaEarth Foundations**
9. **SatMAE**
10. **SatMAE++**
11. **AnySat**
12. **ChangeStar**
13. **BIT**
14. **ChangeFormer**
15. **Siamese/Transformer change-detection families**
16. **RemoteReasoner**
17. **Earth-OneVision**
18. **RSThinker**

Potential SatQuery abilities

2023 image

+

2024 image

+

2025 image

↓

Temporal reasoning

↓

"Which buildings appeared?"

or:

"Which agricultural fields changed?"

or:

"Show the progression of the flood."



7. Retrieval / Vision-Language Representation Models

These are easy to overlook, but potentially extremely valuable.

They answer:

"Which satellite images are semantically similar to this query?"

Important models

1. RemoteCLIP
2. GeoRSCLIP
3. RS5M
4. RS-CLIP variants
5. CROMA
6. SatCLIP
7. GeoCLIP
8. SkyCLIP
9. EarthCLIP-type models
10. Prithvi representations
11. AlphaEarth embeddings
12. TerraMind embeddings

SatQuery possibility

User:

"Find satellite images showing dense industrial development near rivers."

System:

text query
↓
embedding
↓
geospatial retrieval
↓
candidate regions
↓
VLM analysis

That is a much more powerful product than simple image upload.



8. Geospatial / Geographic Foundation Models

These models are not necessarily RS-VLMs.

But they matter because **Earth is geographic**, not merely visual.

Models/families to study

1. AlphaEarth Foundations
2. GeoCLIP
3. SatCLIP
4. GeoRSCLIP
5. RemoteCLIP
6. Clay
7. Prithvi
8. DOFA
9. Tessera
10. SatlasPretrain
11. AnySat
12. CROMA
13. RingMo
14. Scale-MAE
15. SatMAE
16. SSL4EO
17. GFM families
18. LHRs-Bot
19. GeoChat + geographic context
20. Earth Engine / geospatial embedding ecosystems

Why this matters

A future SatQuery could reason about:

location + imagery + geographic context

rather than merely:

pixels.



9. Reasoning / Agentic RS Models

This is probably the **most strategically important category**.

Study these deeply

1. RemoteReasoner
2. RSThinker
3. GeoVLM-R1
4. Earth-OneVision
5. MIGRANT
6. Think2Seg-RS
7. TerraMind
8. EarthDial
9. GeoGround
10. SATGround
11. VLM-R1-type approaches
12. RL-trained geospatial VLMs
13. Tool-augmented RS-VLM systems
14. Active-perception RS-VLMs

A current 2026 RS-VQA comparison already shows newer general models such as **GeoLLaVA-8K, ZoomEarth, SkySenseGPT, GeoVLM-R1 and EarthDial** alongside general-purpose models, demonstrating that the field is moving well beyond the original GeoChat generation.



10. Active Perception / "LOOK AGAIN" Models

This is a newer direction that I would definitely investigate for SatQuery.

Instead of:

image
↓
answer

the model can decide:

image

↓

"I need more detail."

↓

crop / zoom / inspect region

↓

reason again

↓

answer

Relevant technologies

1. **Crop-tool augmented RS-VQA**
2. **Active perception RS systems**
3. **ZoomEarth**
4. **high-resolution VLM pipelines**
5. **SAM/SAM2 interactive perception**
6. **Grounding DINO**
7. **RemoteReasoner**
8. **RSThinker**
9. **VLM-R1 reasoning systems**

This matters because satellite images frequently contain **tiny objects inside huge scenes**. A recent 2026 study explicitly explores crop-tool-augmented active perception for high-resolution RS VQA.



THE MASTER SHORTLIST

If I were building your **SatQuery research database**, I would put these into the first wave.



Priority A — Deep reverse-engineering

RS-VLM / MLLM

1. **Earth-OneVision**
2. **RemoteReasoner**
3. **TerraMind**
4. **EarthGPT**

5. **EarthGPT-X**
6. **SkyEyeGPT**
7. **VHM**
8. **GeoChat**
9. **LHRS-Bot**
10. **SkySenseGPT**
11. **GeoLLaVA-8K**
12. **EarthDial**
13. **RSThinker**
14. **GeoVLM-R1**

Grounding

15. **GeoGround**
16. **SATGround**
17. **SkyEyeGPT grounding**
18. **Earth-OneVision grounding**
19. **RSVG specialist SOTA models**

Segmentation

20. **Think2Seg-RS**
21. **GeoGround**
22. **Popeye**
23. **GeoPix**
24. **Grounded-SAM**
25. **SAM 2 RS pipelines**

EO Foundation Models

26. **SkySense**
27. **SkySense++**
28. **SkySense V2**
29. **Prithvi-EO**
30. **Prithvi-EO-2.0**
31. **TerraMind**
32. **AlphaEarth Foundations**
33. **DOFA**
34. **Clay**
35. **SatlasPretrain**
36. **SatMAE**
37. **Scale-MAE**
38. **Tessera**
39. **AnySat**

- 40. CROMA
- 41. RingMo
- 42. SSL4EO

Retrieval / CLIP

- 43. RemoteCLIP
- 44. GeoRSCLIP
- 45. SatCLIP
- 46. GeoCLIP
- 47. SkyCLIP

Temporal / multi-image

- 48. MIGRANT
- 49. Prithvi-EO-2.0
- 50. SkySense++
- 51. SkySense V2
- 52. ChangeFormer
- 53. ChangeStar
- 54. BIT
- 55. AnySat
- 56. TerraMind temporal capabilities

If we rank by "what SatQuery can learn"

This is more useful than ranking by raw benchmark score.

Rank	Model	What we should steal/learn
	RemoteReasoner	Autonomous geospatial reasoning
	Earth-OneVision	Broad multimodal RS intelligence
	TerraMind	Cross-modal Earth reasoning
4	SkySense V2	Unified multimodal backbone

5	SkySense++	Generalization/few-shot EO
6	MIGRANT	Multi-image reasoning
7	GeoGround	Precise spatial grounding
8	Think2Seg-RS	Reasoning → segmentation
9	VHM	Honesty / negative evidence
10	EarthGPT-X	Sensor fusion
11	Prithvi-EO-2.0	Temporal + multispectral representation
12	AlphaEarth	Planet-scale embeddings
13	SATGround	Structured spatial localization
14	LHRS-Bot	Geographic knowledge integration
15	SkyEyeGPT	Multitask instruction tuning
16	GeoChat	Conversational grounding
17	RemoteCLIP	RS text-image retrieval
18	DOFA	Flexible multispectral representation
19	Clay	General EO foundation representation
20	SatlasPretrain	Large-scale EO pretraining



One VERY important distinction

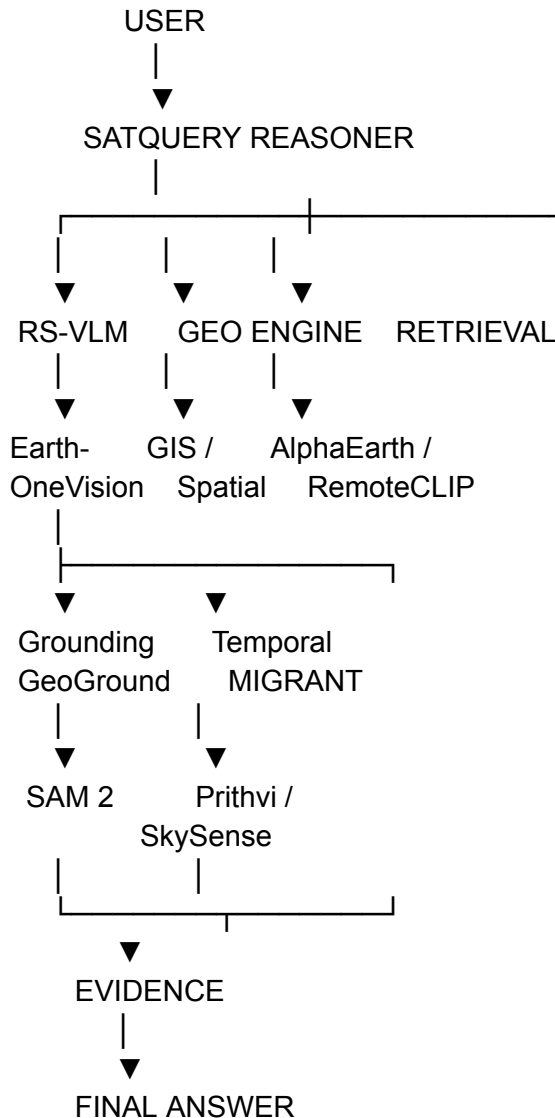
Don't conclude:

Earth-OneVision = best model, therefore use Earth-OneVision.

That's not the correct engineering conclusion.

For example, current benchmark evidence shows that **general RS-MLLMs can be beaten by specialists on individual capabilities**. On the RSVG benchmark, a specialist reaches 84.48 P@0.5 while Earth-OneVision is 65.06; meanwhile on VRSBench, older models such as GeoChat remain competitive on some individual tasks.

So our eventual architecture may look more like:



This is not the final SatQuery architecture. It is the kind of architecture we should evaluate after the model autopsy.



And I would add 5 capabilities to our research checklist

These weren't emphasized enough in your original list, but they could become **major differentiators**:

1. Active perception

"I will zoom/crop and inspect before answering."

2. Temporal intelligence

"I understand what changed, not just what exists."

3. Geospatial computation

"I can calculate area, distance, proximity, intersection, density, etc."

4. Evidence + uncertainty

"Here is what I observed, where I observed it, how confident I am, and what I cannot establish."

5. Autonomous tool selection

"I decide whether this question needs VLM, grounding, segmentation, temporal analysis, GIS, retrieval, or another specialist."

That fifth one is especially important because it turns SatQuery from a **model** into a **system**.

The next thing we should do

Don't jump to architecture yet.

The correct next deliverable is a **massive Model-by-Model Autopsy Matrix** containing roughly **50 serious models/systems** with columns for:

Parameters → **architecture** → **backbone** → **modalities** → **image resolution** → **temporal support** → **VQA** → **captioning** → **counting** → **grounding** → **HBB** → **OBB** → **polygon** → **segmentation** → **change detection** → **SAR** → **multispectral** → **IR** → **video** → **multi-image** → **GIS** → **geographic context** → **reasoning** → **agentic behavior** → **hallucination handling** → **active perception** → **datasets** → **training size** → **benchmark scores** → **GPU/compute** → **inference requirements** → **open-source status** → **license** → **checkpoint availability** → **strengths** → **weaknesses** → **unique innovation** → **SatQuery usefulness**.

That will give us the **actual evidence base** from which we can derive SatQuery's capabilities, instead of guessing features.

And importantly, we should include **both generalists and specialists**—because the current evidence clearly shows that a general RS-VLM can be excellent overall while a specialist can still dominate a particular capability.